

AMANA SAMSUDEEN

Full Stack Software Engineer

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PROFILE

Full-stack software engineer with 2 years of experience designing and shipping production web applications across React, Angular, and Python (FastAPI). Recent work centers on AI-powered systems, building RAG pipelines, integrating LLMs, and turning ambiguous product requirements into deployed, user-facing features. Comfortable owning a feature end to end, from API design through frontend implementation and CI/CD deployment, within Agile teams. Currently completing an MSc in Information Technology.

PROFESSIONAL EXPERIENCE

Software Engineer - X-Venture Pvt Ltd

2023 – 2024 | Colombo, Sri Lanka

- Developed and implemented a module integration feature connecting the application with third-party platforms (Bitbucket, GitHub, and GitLab), enabling code to be pushed directly from the application to these repositories.
- Implemented unit testing frameworks, achieving measurable test coverage improvements that strengthened long-term maintainability and reduced regression risk.
- Actively participated in Agile sprints, sprint planning, daily stand-ups, and retrospectives contributing to on-time delivery of quarterly feature milestones.

Software Engineer Intern - X-Venture Pvt Ltd

2022 – 2023 | Colombo, Sri Lanka

- Fixed 50+ frontend bugs across legacy Angular modules, improving stability and reducing reported UI issues in production releases.
- Collaborated with cross-functional teams to align feature development with client requirements, gaining hands-on Agile Scrum experience.

TECHNICAL SKILLS

Programming Languages: Python, Java, JavaScript, TypeScript, SQL

Frontend Development: React.js, Angular, HTML5, CSS3, Tailwind CSS, Material Tailwind

Backend Development: FastAPI, .NET, Node.js, Express.js, REST API

Databases: PostgreSQL, MongoDB, Firebase, Pinecone Vector DB, SQL Server

AI / Machine Learning: LangChain, RAG Architecture, LLM Integration, Prompt Design

Tools & Platforms: Git, GitHub, Postman, Jira, Vercel, Railway, Google Cloud, VS Code, GoDaddy, Resend, AWS

PROJECTS

ResilienceQ - AI-Powered Resilience Assessment Platform

- Built a fullstack resilience assessment platform (React/Astro + FastAPI + PostgreSQL) where users complete adaptive questionnaires and receive LLM-generated, personalized coaching recommendations.
- Designed a RAG pipeline with Pinecone vector DB and Gemini 2.5 Flash to power a contextual chatbot, improving answer relevance over keyword search by grounding responses in domain-specific content.
- Deployed to production via Railway (backend) and Vercel (frontend) with environment-based CI/CD configuration, achieving reliable uptime for pilot users.

Technologies: React.js, Astro, Material Tailwind, FastAPI, PostgreSQL, Pinecone Vector DB, Gemini 2.5 Flash, LangChain

XAPI - API Management Platform

- Built and maintained frontend modules for API management workflows, implementing responsive UI components and API integration features used by internal teams.
- Collaborated with Spring Boot backend teams to optimize platform data flow, reducing frontend-backend round trips through efficient API contract design.

- Responsible for an integration feature connecting the platform with third-party applications (GitHub, Bitbucket, and GitLab), enabling seamless interaction between the platform and external repositories.

Technologies: Angular, Spring Boot, MongoDB, RESTful API

SMECO - Loyalty Management System

- Architected a fullstack loyalty platform (React.js + Node/Express REST API + MySQL) supporting customer records with points tracking, tier management, and transaction history.
- Used Sequelize ORM with MySQL, implementing cascading transactions to prevent point inconsistencies on failed redemptions and maintain data integrity.

Technologies: React.js, Node.js, Express.js, Sequelize ORM, MySQL

ShieldCare - Parental Control Application

- Developed a cross-platform parental control app with React.js + FastAPI + Firebase, enabling real-time screen time tracking and per-app usage limits configurable from a parent dashboard.
- Used Firebase Realtime Database listeners to push live usage updates to the parent dashboard.

Technologies: React.js, FastAPI, Firebase

PETSO - Social Platform for Animal Welfare Organizations

- Collaborated in a team to build a full-stack social platform (PHP, MySQL) connecting animal welfare organizations with the public, covering adoption listings, organization profiles, and community engagement features.
- Built adoption listing and matching functionality, enabling shelters to publish animal profiles and prospective adopters to search and filter by criteria such as species, age, and location.
- Implemented responsive UI with HTML and SCSS, and deployed the application on AWS.

Technologies: PHP, SCSS, HTML, MySQL, AWS

EDUCATION

MSc in Information Technology - Bharathiar University

2024 – 2026 | Coimbatore, India

BSc in Information Systems - University of Colombo School of Computing

2020 – 2023 | Colombo, Sri Lanka

ICT Technician (NVQ Level 4) - Technical College Batticaloa

2019 | Batticaloa, Sri Lanka

CERTIFICATIONS

- Postman API Student Expert
- Java Full Stack with React.js and AI
- Prompt Design in Vertex AI
- Introduction to Internet of Things
- Computer Networks and Internet Protocol

AWARDS

- ICCR Scholar - Merit scholarship awarded by the Government of India to pursue MSc.
- Winner, PearlHack 1.0 - Inter-university Hackathon, Sabaragamuwa University, Sri Lanka.
- Runner-up, Hackaholics 5.0 - Inter-university Girls' Hackathon, University of Colombo.

PUBLICATIONS

“E-Waste Awareness, Disposal Behaviors and the Path Toward Sustainable Management - A Study Among Undergraduate Students at University of Colombo, Sri Lanka.”

Book chapter in Fuels for Sustainable Transportation, Taylor & Francis Group / CRC Press - ISBN 9781003534679 (published 21 November 2025).